

Question

Given $E^\circ(\text{Pb}^{2+}/\text{Pb}) = -0.1266 \text{ V}$, $E^\circ(\text{PbO}_2/\text{Pb}^{2+}) = 1.458 \text{ V}$, $K_{sp}(\text{PbSO}_4) = 1.8 \times 10^{-8}$,
 $K_{a2}(\text{H}_2\text{SO}_4) = 1.0 \times 10^{-2}$

Calculate the standard electromotive force of a lead-acid battery and select the closest option.

A. 1.5 V

B. 1.6 V

C. 1.7 V

D. 1.8 V

E. 1.9 V

F. 2.0 V

G. 2.1 V

H. 2.2 V

Answer

Correct Answer: **E**

Detailed Explanation

This is a physical chemistry problem, examining the calculation of the Nernst equation in electrochemistry.

First, write out the half-reactions at the positive and negative electrodes:

Negative electrode: $\text{Pb (s)} + \text{HSO}_4^- (\text{aq}) \rightarrow \text{PbSO}_4 (\text{s}) + \text{H}^+ (\text{aq}) + 2\text{e}^-$

Positive electrode: $\text{PbO}_2 (\text{s}) + \text{HSO}_4^- (\text{aq}) + 3\text{H}^+ (\text{aq}) + 2\text{e}^- \rightarrow \text{PbSO}_4 (\text{s}) + 2\text{H}_2\text{O (l)}$

CHECKPOINT

0.5 PTS

Negative electrode: $\text{Pb (s)} + \text{HSO}_4^- (\text{aq}) \rightarrow \text{PbSO}_4 (\text{s}) + \text{H}^+ (\text{aq}) + 2\text{e}^-$

CHECKPOINT

0.5 PTS

Positive electrode: $\text{PbO}_2 (\text{s}) + \text{HSO}_4^- (\text{aq}) + 3\text{H}^+ (\text{aq}) + 2\text{e}^- \rightarrow \text{PbSO}_4 (\text{s}) + 2\text{H}_2\text{O (l)}$

First, discuss the negative electrode. Given $E^\circ(\text{Pb}^{2+}/\text{Pb}) = -0.1266 \text{ V}$, corresponding to the half-reaction equation $\text{Pb}^{2+} (\text{aq}) + 2\text{e}^- = \text{Pb (s)}$, then $E^\circ(\text{PbSO}_4/\text{Pb}) = E^\circ(\text{Pb}^{2+}/\text{Pb}) + \frac{RT}{2F} \ln[\text{Pb}^{2+}]$

Now solve for $[\text{Pb}^{2+}]$:

$$[\text{Pb}^{2+}] = \frac{K_{sp}(\text{PbSO}_4)}{[\text{SO}_4^{2-}]}$$

Also, $[\text{SO}_4^{2-}] = \frac{K_{a2}[\text{HSO}_4^-]}{[\text{H}^+]} = 1.0 \times 10^{-2} \text{ mol/L}$

CHECKPOINT

1 PTS

$$[\text{SO}_4^{2-}] = 1.0 \times 10^{-2} \text{ mol/L}$$

Substitute into $[\text{Pb}^{2+}] = \frac{K_{sp}(\text{PbSO}_4)}{[\text{SO}_4^{2-}]} = \frac{1.8 \times 10^{-8}}{1.0 \times 10^{-2}} = 1.8 \times 10^{-6} \text{ mol/L}$

CHECKPOINT

1 PTS

$$[\text{Pb}^{2+}] = 1.8 \times 10^{-6} \text{ mol/L}$$

Then $E^\circ(\text{PbSO}_4/\text{Pb}) = E^\circ(\text{Pb}^{2+}/\text{Pb}) + \frac{RT}{nF} \ln[\text{Pb}^{2+}] = -0.296 \text{ V}$ (substituting $T = 298\text{K}$, the same applies below)

CHECKPOINT

1 PTS

$$E^\circ(\text{PbSO}_4/\text{Pb}) = -0.296 \text{ V}$$

Now discuss the positive electrode. Given $E^\circ(\text{PbO}_2/\text{Pb}^{2+}) = 1.458 \text{ V}$, corresponding to the half-reaction equation

$$\text{PbO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- = \text{Pb}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}),$$

then

$$E^\circ(\text{PbO}_2/\text{PbSO}_4) = E^\circ(\text{PbO}_2/\text{Pb}^{2+}) + \frac{RT}{2F} \ln \frac{[\text{H}^+]^4}{[\text{Pb}^{2+}]}$$

Similarly, $[\text{Pb}^{2+}] = 1.8 \times 10^{-6} \text{ mol/L}$

Substitute into $E^\circ(\text{PbO}_2/\text{PbSO}_4) = E^\circ(\text{PbO}_2/\text{Pb}^{2+}) + \frac{RT}{2F} \ln \frac{[\text{H}^+]^4}{[\text{Pb}^{2+}]} = 1.62 \text{ V}$

CHECKPOINT

1 PTS

$$E^\circ(\text{PbO}_2/\text{PbSO}_4) = 1.61 \text{ V}$$

Therefore, the electromotive force is $1.62 - (-0.296) = 1.92 \text{ (V)}$

CHECKPOINT

0.5 PTS

The electromotive force is closer to 1.9 V